

Solid biofuel generation from Eucalyptus wood residues: An experimental comparison study of torrefaction, hydrothermal, and slow pyrolysis

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ABSTRACT

Eucalyptus trees are extensively cultivated to meet paper pulp, timber, and fiber-demands, generating substantial amounts of wood residues (EWR), suitable for energy applications. This study evaluates the conversion of EWR via torrefaction (TOR), hydrothermal carbonization (HTC), and slow pyrolysis (SP) in lab-scale reactors, assessing temperature and time effects on fuel properties. The resulting solid fuel was characterized using proximate and elemental analyses as well as thermogravimetric analysis. Results indicate that product yields were significantly influenced by both temperature and residence time. The highest temperatures within the investigated operating ranges produced the most pronounced fuel upgrading, with volatile matter decreasing from 86.9% to 78.9%, 45.1%, and 14.6% for TOR, HTC, and SP, respectively, while their higher heating value increased to approximately 22, 26 and 33 MJ/kg. A central composite design experiment was implemented, with statistical analyses conducted on the resultant data, employing response surface methodology and analysis of variance. Under optimal condition of 194.32°C-14.59 min (TOR), 180.57°C-2.38 h (HTC), and 282.85°C-3.09 h (SP), energy yields of 97%, 84%, and 71% were obtained, respectively. The solid fuels showed combustion behavior comparable to coal and lignite. Emission estimations suggested lower environmental impact than fossil coal, reinforcing their potential as sustainable biofuels.

Nomenclature

Abbreviations		MC	Moisture content
AC	Ash content	MY	Mass yield
CCD	Central composite design	NO _x	Nitrogen oxides
CH ₄	Methane	RSM	Response surface methodology
CO	Carbon monoxide	SBC	Specific bioexergy content
CO ₂	Carbon dioxide	SP	Slow pyrolysis
db	Dry basis	T	Temperature
DTG	Derivative thermogravimetric	TGA	Thermogravimetric analysis
EDF	Energy densification factor	TOR	Torrefaction
EF	Emission Factor	VM	Volatile matter
EWR	Eucalyptus wood residues		
EY	Energy yield	Units	
FC	Fixed carbon	°C	Celsius degrees
FR	Fuel ratio	kg	Kilogram
GHG	Greenhouse gases	MJ	Megajoule
HHV	Higher heating value	min	Minute

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HTC	Hydrothermal carbonization	wt. %	Weight percentage
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1. Introduction

The sustainable production of bio-based fuels and chemicals requires feedstocks that do not compete with food production or drive land-use changes associated with dedicated energy plantations. Lignocellulosic residues have therefore gained attention as alternative resources capable of supporting bioenergy generation while mitigating food-security concerns. In Brazil, for instance, the forest industry represents as major source of such residues due to the extensive cultivation of *Eucalyptus* species, which accounted for approximately 77% of the country's 10.52 million hectares of planted forest area in 2024 [1]. During harvesting operations, a significant fraction of tree biomass remains as

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residue. While some studies estimate that residues represent approximately 12% of the total tree biomass [2] others report substantially higher shares, ranging from 30 to 40% of the harvested material [3], depending on assessment methodology.

In Brazil, *Eucalyptus* plantation primarily supports the pulp and paper industry, although part of the harvested wood is used for engineered wood products and charcoal production. The country ranks among the world's leading producers of pulp, paper, and charcoal, with most charcoal derived from cultivated *Eucalyptus* plantations [4]. As a result, large quantities of *Eucalyptus* wood residues (EWR) are generated during harvesting and processing, including sawdust, bark, fines, and yard residues. Despite their availability, these residues are often underutilized, sometimes leading to accumulation or uncontrolled burning. Their valorization through biomass-fired systems or co-firing applications have been proposed as a potential valorization pathway that may reduce greenhouse gas emissions and improve resource efficiency.

However, the direct use of raw biomass presents limitations, including low energy density, high moisture content, susceptibility to biological degradation, and limited grindability. These properties hinder storage, transportation, and direct substitution for fossil coal in pulverized systems. Thermochemical conversion processes can significantly improve fuel characteristics by producing more stable, energy-dense solids with combustion behaviors closer to coal. Table 1 compares key properties of raw lignocellulosic biomass with those obtained after thermochemical treatments.

Several thermochemically technologies are available for upgrading biomass into solid energy carriers, among which slow pyrolysis (SP), torrefaction (TOR), and hydrothermal carbonization (HTC) are the most extensively studied.

Slow pyrolysis (SP), involves the thermal degradation of biomass under inert conditions at temperatures typically between 300 and 800 °C and relatively low heating rates, aiming to maximize biochar production [13]. Increasing temperature promotes devolatilization and thermal cracking of structural polymers, reducing solid yield while enriching the remaining char in fixed carbon [13]. In Brazil, SP is widely used for large-scale biochar production due to its relatively low energy demand and investment costs, although much of the domestic production still relies on rudimentary kilns with limited process control [14].

Torrefaction is a milder process conducted at 200 and 300 °C under inert or oxygen-limited atmospheres [13]. It primarily induces partial devolatilization and hemicellulose degradation, reducing moisture and oxygen-rich functional groups while enhancing hydrophobicity, grindability and calorific value [5]. Product properties depend on reactor design, temperature, residence time, and heat-transfer characteristics, which influence devolatilization rates and energy densification [7]. For *Eucalyptus*, TOR can yield solids with higher heating values (HHV) of up to 23 MJ/kg [15]. Although fuel upgrading is less extensive than in SP, TOR benefits from shorter residence times and relatively high energy yields. However, despite its technical maturity, large-scale commercialization remains limited [16].

Hydrothermal carbonization (HTC) differs fundamentally from dry

thermochemical routes by operating in an aqueous medium between 180 and 260 °C under autogenous pressure [8]. Its main advantage lies in processing without prior drying. Under hydrothermal conditions, biomass undergoes a sequence of reactions including hydrolysis, dehydration, decarboxylation, and condensation reactions, forming a carbon-rich hydrochar while soluble intermediates partition into the liquid phase and CO₂ is released as the primary gaseous product [17]. Compared to dry routes like pyrolysis and gasification, HTC operates at lower temperatures and typically retains 50–90% of the initial mass and 80–95% of the initial energy in the solid fraction [17]. Hydrochar often exhibits reduced inorganic content relative to the raw biomass, potentially mitigating ash-related issues if combusted [8]. Hydrochar applications include solid fuel use, absorption, soil amendment, and carbon sequestration [13].

Despite numerous studies have investigated individual thermochemical conversion routes, direct and systematic comparisons of the solid fuel properties obtained from different pretreatment technologies are still limited, with only a few representative studies available [18–20]. While previous works have examined the conversion of EWR via TOR, SP, or HTC separately, a systematic experimental comparison of these three thermochemical pathways applied to this specific biomass under a unified framework has not yet been reported. This gap hinders a systematic evaluation of how key operating parameters, particularly temperature and residence time, influence solid product characteristics across different conversion technologies. In this context, the novelty of the present study lies in providing a unified experimental assessment of solid fuels derived from *Eucalyptus* wood residues through TOR, SP, and HTC, combined with a quantitative evaluation of the effects of operating conditions using a central composite design (CCD) and response surface methodology (RSM).

Beyond a simple performance comparison, the approach adopted in the current study enables the identification of optimal operating conditions to maximize energy yields, a consistent comparison of energy densification behavior, an analysis of the combustion performance of the derived solid fuels, and an estimation of the theoretical emissions associated with their combustion, thereby offering a comprehensive and comparative interpretation of conversion behavior across the three thermochemical routes.

2. Materials and methods

2.1. Materials

Eucalyptus wood residues were collected from the experimental units of the Federal University of Viçosa, Minas Gerais state, Brazil. Samples were ground in an electric hammer mill, with a fixed speed of 3520 rpm, linked to a sieve with a mesh size of 1 cm. The grinding process was necessary to achieve a consistent size distribution of the feedstock particles. After grinding, the biomasses were dried at 105 ± 2 °C for 24 h and subsequently storage in airtight polyethylene bags at ambient temperature.

Table 1
General characteristics of raw and thermochemically treated lignocellulosic biomass compared to coal.

Properties	Unit	Wood chips	Torrefied wood	Hydro-treated wood	Pyrolyzed wood	Bituminous coal
MC	%wt	20–60*	3.0*	Saturated	8 ^φ	5–10*
VM	%wt	70–85 ^γ	61–82 ^α	48–79 ^γ	9–17 ^φ	-
Ash	%wt	0.7–2**	<3.0**	0.1–0.2 ^γ	3–22 ^φ	10–40**
LHV	MJ/kg	6–13*	19–22*	20.4–24.2 ^γ	31.93 ^β	>25*
EDF	-	0.7–0.9*	1.3*	1.1–1.3 ^γ	1.6–2.5 ^φ	5.6–7*
Mass yield	%	-	32–96 ^α	51.6–72.3 ^γ	30 ^β	-
Hygroscopic nature		Hydrophilic***	Moderately hydrophobic***	Moderately hydrophobic ^γ	Hydrophobic [†]	Hydrophobic***
Biological degradation		Fast***	Slow***	Slow after drying ^γ	Slow [†]	None***
Product consistency		Limited***	High***	High ^γ	High [†]	High***

Adopted from Ref. [5]* [6], ** [7], *** [8], ^γ [9], ^β [10], ^α [11], ^φ and [12][†]. MC-moisture content; VM-volatile matter; LHV-low heating value; EDF-energy densification factor.

2.2. Experimental procedure and design

The schematic representation of torrefaction, hydrothermal carbonization, and slow pyrolysis are shown in Fig. 1. Detailed information related to the experimental setups and procedures is provided in Text S1 (Supplementary Material).

The design of experiments (DOE) allows to evaluate the combined effect of temperature and time experiments by the response surface method on the thermochemical experimental treatments. A central composite design (CCD) was used for the DOE. A total number of 11 experiments were conducted. Table 2 shows the summary of experimental ranges of independent variables. Each treatment was tested in duplicate. As the design considered two levels for each continuous factor, a quadratic regression model was used. The significance of each term was evaluated based on analysis of variance (ANOVA) at significance level $\alpha = 0.05$.

2.3. Analytical methods

2.3.1. Proximate, ultimate, and heating value analysis

The EWR and the solid product derived from the thermochemical conversion were characterized following the same methodology described by the authors in previous publications [21]. Proximate analysis was conducted to determine moisture (MC), volatile matter (VM), and ash contents (AC), while fixed carbon (FC) was calculated by difference. Higher heating values were measured using a bomb calorimeter. Ultimate analysis was performed to determine carbon (C), hydrogen (H), and nitrogen (N) contents, and oxygen (O) was calculated by difference on a dry-ash-free basis. All measurements were performed in duplicate and average values are reported.

To evaluate process performance, solid mass yield (MY), energy densification factor (EDF), energy yield (EY), and fuel ratio (FR) were calculated. The corresponding equations and detailed definitions are provided in Text S2 (Supplementary Material).

2.3.2. Thermogravimetric analysis

Thermogravimetric analysis (TGA) was performed with an STA 449C thermogravimetric analyzer (Netzsch Instruments, Germany). For the analysis, about 10 mg of the sample were placed inside Al_2O_3 sample holder, the furnace was sealed and purged with high purity (99.9995 %) nitrogen flow to remove air. The sample is heated at a rate of 20 °C/min from room temperature to 900 °C. Compressed air (oxidizing

Table 2

Experimental range and values of independent variables used in central composite design.

Independent variable	$-\alpha$	-1	0	1	$+\alpha$
Torrefaction					
Temperature [°C]	179.29	200	250	300	320.71
Residence time [min]	12	15	22.5	30	33
Hydrothermal Carbonization					
Temperature [°C]	168	180	210	240	252
Residence time [hour]	2.5	3	4.5	6	6.5
Slow pyrolysis					
Temperature [°C]	238	300	450	600	662
Residence time [hour]	3	3.5	4.5	5.5	6

atmosphere) was used to perform the thermogravimetric test to ensure combustion conditions, at a constant gas flow rate of 250 ml/min. The ignition temperature (T_i), and burnout temperature (T_b), were determined according to Ref. [22]. The temperature at which the maximum derivative thermogravimetric (DTG) value takes place is defined as the peak temperature (T_p).

2.3.3. Emissions assessment

The estimation of emission factors (EF) serves as a useful tool for a preliminary assessment of potential emission from biomass, biochar and hydrochar if used for energy purposes. Helpful correlations can be formulated to estimate the characteristics of gaseous emissions such as CO , CO_2 , CH_4 , and NO_x , based on the emission factor approach [22]. Therefore, to estimate the potential combustion-related emissions of the samples studied, the theoretical emissions factors (in kg per ton) were calculated using their ultimate analysis. The equations used for the determinations of the emission factors are described in Text S3 (Supplementary Material).

3. Results and discussion

3.1. Mass and energy yield of biochar and hydrochar

The proximate analysis, atomic ratios and calorific values of the *Eucalyptus* feedstock, and derived solid products are presented in Table S1 (Supplementary Material). The experiments evaluate the influence of temperature and residence time on the solid obtained from TOR, HTC, and SP. To ensure reproducibility, the center point in the

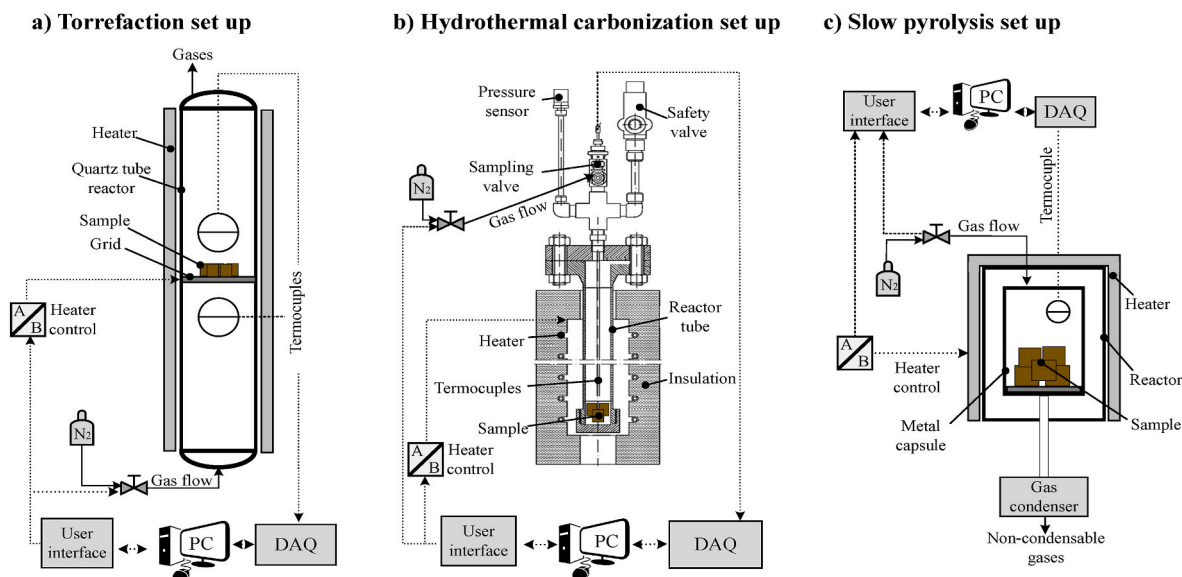


Fig. 1. Experimental setup for torrefaction (a), hydrothermal carbonization (b), and slow pyrolysis (c).

CCD was repeated three times.

All conversion routes produced solid products with distinct yields and properties. The ANOVA analysis results (Supplementary Material) confirmed temperature as the dominant parameter influencing product characteristics. As shown in Fig. 2, increasing process severity (combined temperature and residence time), reduce solid yield, with ranges of 73.25–92.69% (TOR), 47.1–72.34% (HTC), and 29.05–53.41% (SP). The decrease in MY reflects progressive devolatilization and structural degradation. In SP and TOR, removal of moisture and volatile together with partial decomposition of hemicelluloses, cellulose and lignin increase relative carbon content. In HTC, the process takes place in a water-rich environment where reactions promote solubilization of organic fractions and partial gas formation (mainly CO₂), further contributing to mass loss [20].

Maximum mass losses were 30% (TOR), 53% (HTC) and 75% (SP). These progressive reduction in yield resulted in increased energy density, and energy and exergy content (HHV and SBC), with HHV ranging from 20.62 to 21.58 MJ/kg (TOR), 20.54–25.62 MJ/kg (HTC), and 26.22–32.51 MJ/kg (SP). Consequently, SP produced the most energy-dense material, followed by HTC and TOR.

Although TOR and HTC operates within similar temperature ranges, HTC resulted in significantly higher mass loss (Figs. 3 and 4). This difference arises from the distinct reaction media: in HTC, water acts both as a reactant and as a catalyst, enhancing biomass breakdown and promoting transfer of soluble and part of the inorganic fraction to the liquid phase [23]. In contrast, TOR largely retains inorganic constituents within the solid matrix, resulting in higher apparent. Similar trends have been reported for lignocellulosic biomass, where MY under identical nominal temperatures was substantially lower in HTC than in TOR [24].

3.2. Process performance modeling

The results listed in Table S1 (Supplementary Material). show that

both temperature and time significantly influence the VM content, which directly impacted MY and sample upgrading. Increasing temperatures promoted devolatilization, reducing VM to 78.92–85.45 wt% (TOR), 45.09–77.99 wt% (HTC), and 13.15–45.05 wt% (SP). This reduction in VM leads to an increase in FC content, as illustrated in Fig. 5, confirming progressive carbonization with increased treatment severity.

For TOR, the highest FC values (~21 wt%). were achieved at 300–320 °C. In contrast, SP, produced biochars exceeding 80 wt% FC at around 600 °C, approaching the range of bituminous coal and anthracite (80–90 wt%) [25]. Such high FC contents are essential for specific applications, such as metallurgy, where biochar must substitute fossil sources with comparable structural and combustions characteristics [26]. These values align with those reported in previous studies on *Eucalyptus* biochar produced under similar operational conditions [27]. However, it should be noted that even when comparable technologies and nominal conditions are used, variations in biochar characteristics can still occur, highlighting the sensitivity of biochar properties to subtle differences in process control and reactor thermal dynamics.

The increase in FC translates into higher fuel ratio (FR). This is a key indicator of combustion suitability. Solids with high FR burn more steadily, and provide a sustained heat output, while low FR values favor easier ignition and faster combustion [28]. FR values rose from 0.15 in raw EWR to 0.26 (TOR), 1.21 (HTC), and over 4.51 (SP). As shown in Fig. 5, SP-derived biochars (450–600 °C) exhibited FR values between 4.5 and 6.5, indicating a likely slow and stable combustion behavior. Such values approach those of metallurgical coke and are considered favorable for iron ore sintering and reductive process [26].

Cai et al. [29] developed a mathematical model evaluating biomass-based fuels substitution in sintering operations, concluding that replacement ratios above 30% may reduce sintering yield due to lower FC and FR relative to coke. Safarian [30]. reported that biochar with FC contents exceeding 80 wt% could replace up to 60% of coke breeze while

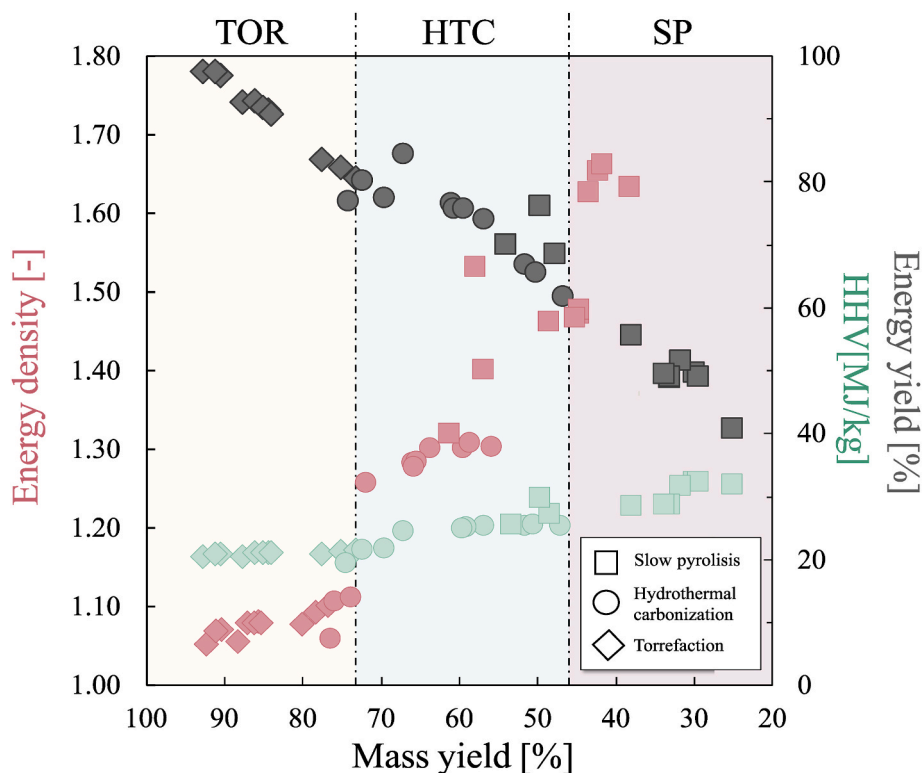


Fig. 2. Relation between heating value, energy yield, energy density and mas yield for biochar and hydrochar. The yellow, blue and red color areas indicate the zones where torrefaction, hydrothermal carbonization and slow pyrolysis mainly take place, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

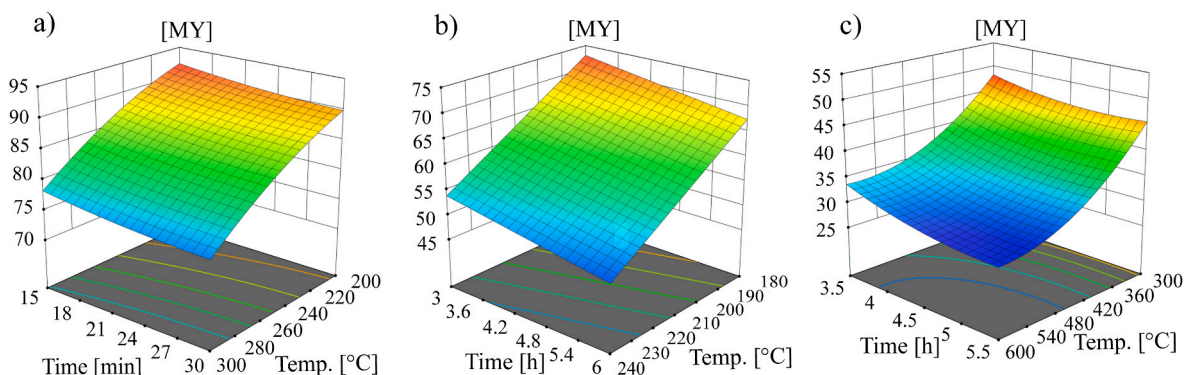


Fig. 3. Mass yield 3D response surface for a) torrefaction, b) hydrothermal carbonizations, and c) slow pyrolysis.

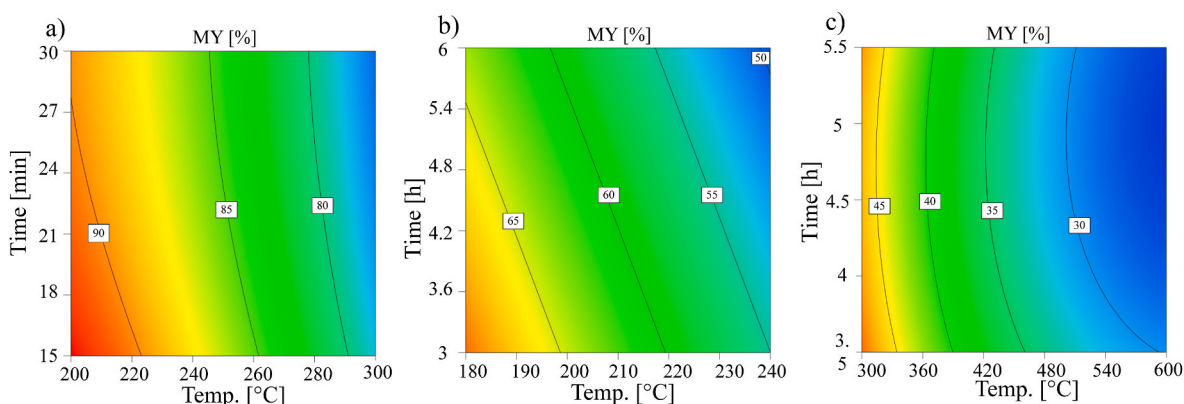


Fig. 4. Mass yield contour plots for a) torrefaction, b) hydrothermal carbonizations, and c) slow pyrolysis.

maintaining sinter yields above 80%, and suggested in mini blast furnaces complete substitution could be technically feasible. The high FR values achieved via SP in the present study therefore position pyrolytic biochar as the most suitable candidate for metallurgical integration among the three routes evaluated.

The compositional changes are further evidence in the VanKrevelen diagram (Fig. 6). All treatments reduced H:C and O:C atomic ratios, following dehydration, decarboxylation and demethylations pathways typical of thermochemical upgrading. These reactions, including glucose degradation and CO₂ release via carboxyl group removal, were mainly observed at higher temperatures [8]. In contrast, at relatively low thermal treatment such as TOR, decarboxylation is not dominant. Instead, the process primarily modifies the physicochemical properties of biomass, improving handling, storage, and combustion characteristics rather than inducing extensive chemical transformation. TOR samples clustered near the biomass region (O:C ~0.56; H:C ~1.42), while HTC samples shifted to intermediate regions (O:C 0.29–0.47; H:C 0.82–1.30), consistent with partial carbonization under hydrothermal conditions. SP biochar exhibited the greatest reduction (H decreased by 37–81%; O by 54–80%), positioning them near lignite, coal, and anthracite region, indicating enhanced aromatic condensation structural stabilization.

Specific bioenergy content (SBC) of the samples (colormap Fig. 6), represents the exergy content of biological materials that can be converted into useful energy [31]. SBC increased with treatment severity ranging from 21.59 to 22.60 MJ/kg (TOR), 21.51–26.80 MJ/kg (HTC), and 26.80–34.08 MJ/kg (SP), falling within previous reported ranges. For instance, Tamires da Silva et al. [32] reported SBC values of hydrochar produced from *Eucalyptus* sawdust, of up to 21.53 MJ/kg at a temperature of 230 °C with a 60-min reaction time. However, the authors also note that the use of a catalyst (H₂SO₄) under the same operational conditions can significantly impact process performance,

resulting in an increase in SBC values of up to 26.43 and 26.93 MJ/kg using 0.1 mol/L and 0.2 mol/L of catalyst, respectively.

3.3. Process optimization

The optimization process was designed to simultaneously adjust the magnitude of input variables to achieve the best overall response when multiple factors influence a set of responses. In previous sections, the impact of temperature and residence time on MY, EY, and HHV was established. Based on these findings, the first optimization strategy focused on maximizing EY while minimizing temperature and residence time MY and HHV were constrained to remain within their obtained limits. The applied constraints are summarized in Table 3, and the corresponding optimal solutions with the highest desirability are presented in Table 4.

Since thermochemical upgrading inherently involves trade-offs, no single response can be maximized independently without affecting the others. For all three conversion routes, the first optimization identified operating conditions near the lower bounds of temperature and residence time. This reflects the strong dependence of EY on solid yield: higher temperatures and longer residence times enhance devolatilization and carbonization but simultaneously reduce mass yield through increased thermal degradation. Consequently, operating at milder conditions favors energy retention despite producing solids with lower HHV. To address this limitation, a second optimization strategy was targeted HHV values close to the average experimental values (21 MJ/kg for TOR, 23 MJ/kg for HTC, and 29 MJ/kg for SP), while maintaining EY as high as possible and achieving a global desirability greater than 0.80. A desirability value within the range of 0.80 to 1.00 is generally considered indicative of an excellent optimization outcome [33].

For TOR, the targeted HHV was approached but not fully achieved

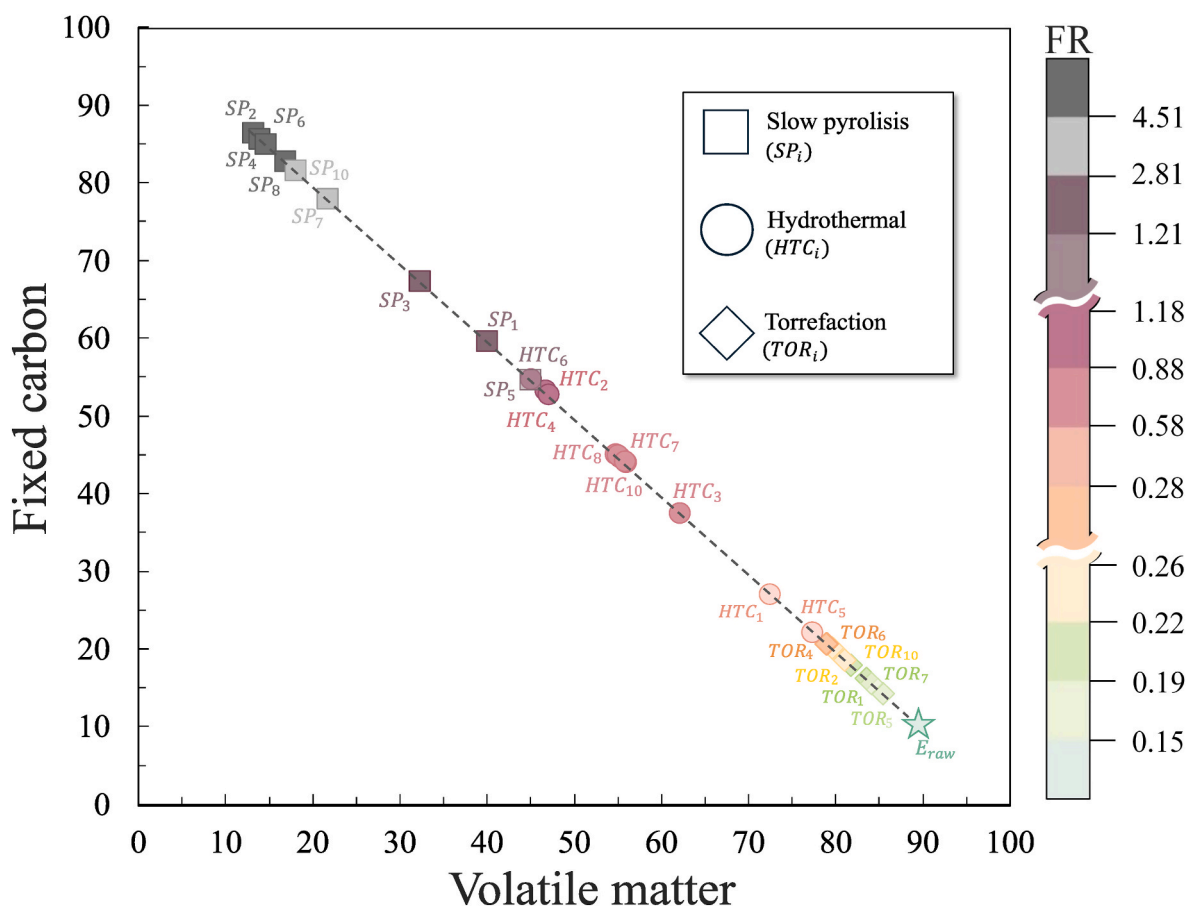


Fig. 5. Correlation between FC and VM of obtained biochar and hydrochar. The subscript *i* refers to the sample run number, as indicated in Table S1 (Supplementary Material).

(20.811 MJ/kg; desirability 0.834). Although slightly below the target, these results indicate that the optimization was able to approach the desired HHV within the limits of the experimental domain, suggesting that the optimization model may have encountered constraints related to the experimental data and process boundaries. In contrast, both HTC and SP successfully achieved the targeted HHV values while maintaining satisfactory desirability scores. In these cases, moderate increase in temperature was sufficient to enhance carbon enrichment without excessively compromising EY.

Optimizing a process, whether prioritizing EY, HHV, a balance of both, or any other parameter, depends primarily on the specific goals, requirements, and constraints of the intended application. In this context, process optimization should not be viewed as a one-size-fits-all solution, but rather as a flexible tool for tailoring operational conditions to specific goals in order to meet diverse technological, economic, and environmental objectives.

3.4. Thermal degradation behavior

The thermogravimetric (TG) and derivative (DTG) curves of raw EWR and selected hydrochar and biochar samples are presented in Fig. 7. These curves provide a comparative visualization of the thermal behavior of the material generated in each conversion treatment. The characteristic parameters reflecting the thermal degradation during combustion process are summarized in Table S2 (Supplementary Material), which also includes reference data for biochar and hydrochar reported in the literature.

The overall degradation can be divided into three mayor stages corresponding to initial mass loss (Stage I), main devolatilization (Stage II), and fixed carbon combustion (Stage III).

Stage I (<200 °C) corresponds to moisture evaporation and partial release of light volatiles, with mass losses between 2 and 5 wt.% and no significant structural decomposition. The main devolatilization phase occurs in Stage II (200–390 °C) primarily associated to hemicellulose decomposition (which mainly takes place at 200–315 °C), followed by cellulose degradation, while lignin contributes to a lesser extent [21].

Untreated EWR exhibited earlier degradation, and lower ignition temperature (T_i), indicating higher reactivity. T_i marks the onset of sample degradation and serves as an indicator related to the likelihood of explosion or fire. In contrast, all treated samples showed higher T_i values, reflecting enhanced structural stability due to reduced H:C and O:C ratios and increased C–C bonding (Section 3.2), demonstrating increases resistance to oxidation.

The DTG curve of raw EWR, displayed a shoulder between 250 and 310 °C, attributed to hemicelluloses decomposition. This feature is absent in torrefied samples, indicating that hemicelluloses were effectively degraded in this treatment. TOR-biochar had its main degradation at a slightly higher rate and at a lower temperature (22.42 %/min at 330 °C) compared to the raw EWR (20.69 %/min at 336 °C), suggesting structural rearrangement and partial cellulose breaking down [10]. Hydrochar exhibited a less pronounced shoulder and a broader degradation profile in Stage II, suggesting partial dissolution and formation of more stable structures as product of further cellulose breakdown. Pyrolytic biochar showed no significant degradation during Stage II; its degradation began only near the upper temperature limit of this stage, reflecting its low volatile content.

Stage III (>400 °C) corresponds to the degradation of high-molecular-weight compounds, char formation and subsequence combustion. In this stage hydrochar and pyrolytic biochar had their main degradation, having higher peak temperatures (T_p) (510 and 607 °C,

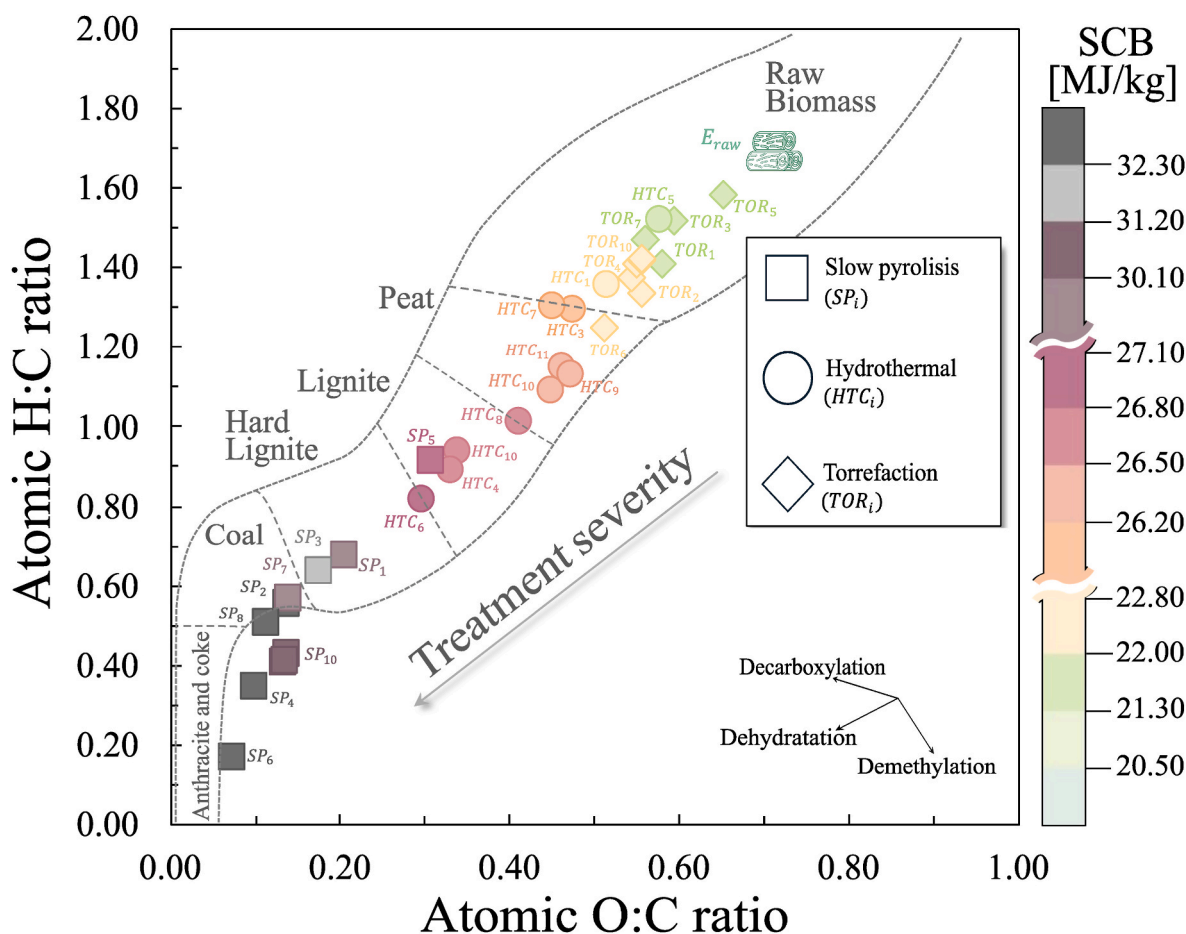


Fig. 6. Van Krevelen diagram for EWR and torrefaction, hydrothermal and pyrolysis experiments.

Table 3
Optimization condition (constraints).

Process	Constrains names	Goal Opt. 1	Goal Opt. 2	Lower limit	Upper limit	Lower weight	Upper weight	Importance
TOR	Temperature [°C]	minimize	minimize	179.30	320.70	1	1	3
	time [min]	minimize	minimize	11.89	33.11	1	1	3
	MY [%]	in range	in range	73.25	92.68	1	1	3
	EY [%]	maximize	maximize	80.80	91.74	1	1	5
	HHV [%]	in range	target	20.62	21.59	1	1	3
HTC	Temperature [°C]	minimize	minimize	167.57	252.43	1	1	3
	time [h]	minimize	minimize	2.38	6.62	1	1	3
	MY [%]	in range	in range	41.10	73.89	1	1	3
	EY [%]	maximize	maximize	61.45	84.61	1	1	5
	HHV [%]	in range	target	20.54	25.62	1	1	3
SP	Temperature [°C]	minimize	minimize	237.87	662.13	1	1	3
	time [h]	minimize	minimize	3.09	5.91	1	1	3
	MY [%]	in range	in range	29.05	53.41	1	1	3
	EY [%]	maximize	maximize	47.85	71.59	1	1	5
	HHV [%]	in range	target	26.22	32.51	1	1	3

respectively). Higher T_p values indicate prolonged combustion. As treatment severity increases, greater degradation of holocelluloses occurs, resulting in a solid fuel enriched in lignin-like compounds within its final carbonaceous structure [32]. The high T_p and degradation rate of SP-biochar, suggest that the aromatic structures within biochar undergo depolymerization, breaking down into smaller and less stable fragments, leading to a rapid increase in degradation rate.

Combustion is considered complete when the burnout temperature

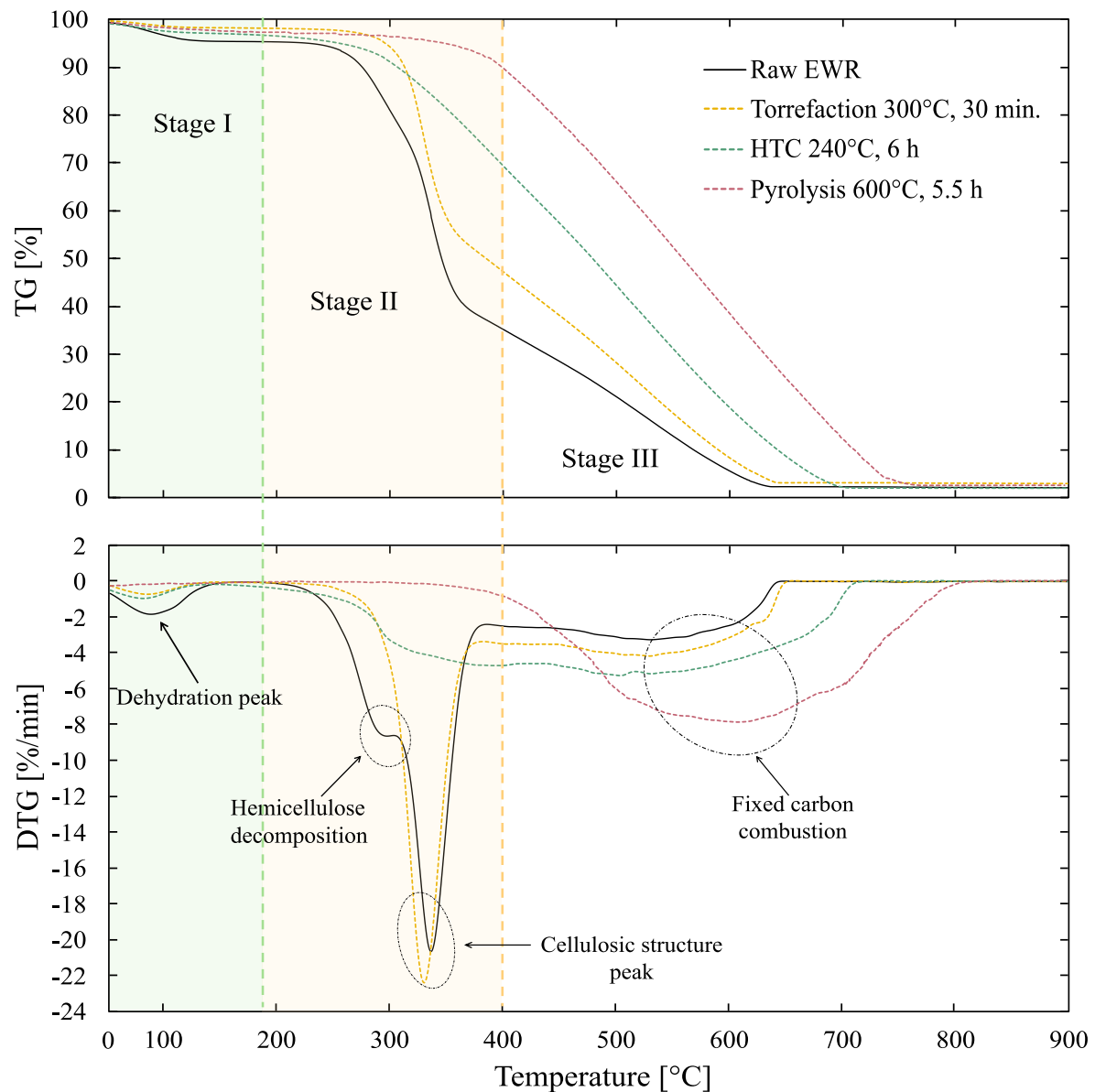
(T_b) is reached. Higher T_b indicates more stable materials that burn more slowly. All treatments increased T_b relative to raw EWR which was 617 °C. TOR and HTC resulted in moderate increases (628 and 679 °C, respectively), whereas SP-biochar exhibited the highest T_b (728 °C), confirming the formation of highly carbonized and thermally stable material.

Comparison with literature data confirms that T_i , T_b , and T_p , are sensitive to TGA operating conditions, particularly heating rate and

Table 4

Desirability values, operational parameters and responses solutions from optimization process.

Process	Temperature	time	MY [%]	EY [%]	HHV [MJ/kg]	Desirability
TOR						
Optimization 1	194.32 °C	14.59 min	92.69	97.71	20.62	0.934
Optimization 2	179.29 °C	19.13 min	92.57	98.12	20.81	0.834
HTC						
Optimization 1	180.57 °C	2.38 h	73.89	84.03	21.69	0.945
Optimization 2	191.29 °C	2.38 h	70.46	82.02	23.00	0.893
SP						
Optimization 1	282.85 °C	3.09 h	51.93	70.21	26.23	0.944
Optimization 2	371.77 °C	3.09 h	43.65	64.21	29.00	0.829

**Fig. 7.** TG and DTG curves for raw EWR and selected biochar and hydrochar samples.

sample mass. As shown in Table S2 (Supplementary Material A), even under a similar atmosphere and heating rates, variability remains due to intrinsic biomass characteristics and equipment configuration. Therefore, T_b , T_p , and T_d should be interpreted as characteristic ranges rather than fixed values, especially when comparing different thermochemical pathways.

3.5. Estimated gaseous emission during combustion

Biomass is widely considered as a renewable energy vector with potential neutral climate impacts due to its close carbon cycle. As result, emissions from biomass combustion are often overlooked when assessing new feedstocks for bioenergy purposes, raising concerns about the true climate impacts [34]. Additionally, from an engineering

perspective, understanding the patterns and characteristics of emissions from biomass combustion is essential for the development and optimization of effective emission control technologies [35]. Therefore, the estimation of gas emissions has a key role providing valuable indicators to evaluate the potential use of biomass and derived biofuels.

Fig. 8 presents the emissions factor (EF) for CO, CO₂, CH₄ and NO_x derived from theoretical estimations presented in Section 2.3. Although this approach does not capture the dynamic evolution of gaseous species during real combustion, it provides a consistent comparative framework for screening-level evaluations of different thermochemically treated fuels. The results should therefore be interpreted as indicative rather than absolute emission predictions.

Samples are arranged in order of increase treatment severity to highlight trends across TOR, HTC, and SP. As a general increase in CO, CO₂, and CH₄ emissions were observed with increasing severity. The CO emission ranged between 61.16 and 68.52, 64.8–83.9 and 82.42–109.98; the CO₂ between 1497.75 and 1677.9, 1586.82–2054.54, and 2018.19–2693.27; and the CH₄ emissions ranged between 2.91 and 3.26, 3.09–3.99 and 3.92–5.24 for TOR, HTC and SP, respectively. In each thermochemical conversion, the sample generated under the most severe operational conditions reflected the highest emission factors for CO, CO₂, and CH₄. For torrefaction, the highest emission factors were observed in the sample processed at ~320 °C for 22 min, while for HTC, the hydrochar produced at ~252 °C for 4.5 h exhibited the highest values. Similarly, for pyrolysis, the biochar obtained at 662 °C for 4.5 h reflected the highest values. This trend reflects the progressive carbon enrichment of the solid fuels: as FC increases, theoretical carbon base emissions per unit mass of fuel also arise. The increase relative to EWR reached approximately 16.9% (TOR), 43.1% (HTC), and 87.6 % (SP), with the highest values corresponding to the most severe operational conditions in each process. These results align with other forest residues, wood waste, and derived-fuels emission factors [22]. These findings indicate that, while thermal treatments enhance biomass fuel properties, they also increase emissions associated with its use as a fuel.

Notably, EF of SP-biochar approached those of medium-to high-rank

coals, as shown in Fig. 8. Biochars produced under mild conditions resemble bituminous coal, whereas those generated at higher temperatures aligned more closely with anthracite. This behavior is also consistent with their position in the Van Krevelen diagram (Section 3.2), where high temperature SP-biochar occupy regions corresponding to anthracite, coke, and hard lignite.

Opposed to carbon emissions, NO_x emissions, did not exhibit a clear monotonic trend. TOR and HTC showed minimal variation (~1–1.3 kg per ton), consistent with the relatively stable nitrogen content in these products. In contrast, nitrogen content slightly increased in SP-biochar as the severity of pyrolysis intensified. As result, SP-biochar produced at more severe conditions showed NO_x emission factor comparable to anthracite, but significantly lower than those of bituminous coal.

4. *Eucalyptus* biochar potential, research limitations, and future perspectives

Biochar represents both, a carbon sequestration material and an upgraded solid fuel. Its long-term stability in soil depends on factors such as feedstock source, and soil conditions. Biochar carbon is estimated to remain in the soil for hundreds to thousands of years, with more than 40% of its carbon still present after 100 years [39]. In comparison, non-carbonized organic materials typically decompose much faster, with soil carbon levels decreasing below 20% within 5–10 years.

This distinction is particularly relevant in Brazil, where large-scale *Eucalyptus* plantations and agricultural expansion create opportunities to couple biomass valorization with long-term carbon management. Moreover, CO₂ released during combustion of eucalyptus-derived biochar can be reabsorbed through sustainable plantation management and continuous replanting. In Brazilian energy plantations, *Eucalyptus* stands are periodically renewed, and part of the residual biomass (e.g., roots, leaves, bark) remains in the soil after harvest, supporting ongoing carbon cycling and contributing to potential long-term sequestration.

Torrefaction, hydrothermal carbonization, and slow pyrolysis have garnered growing interest for the generation of biochar from different

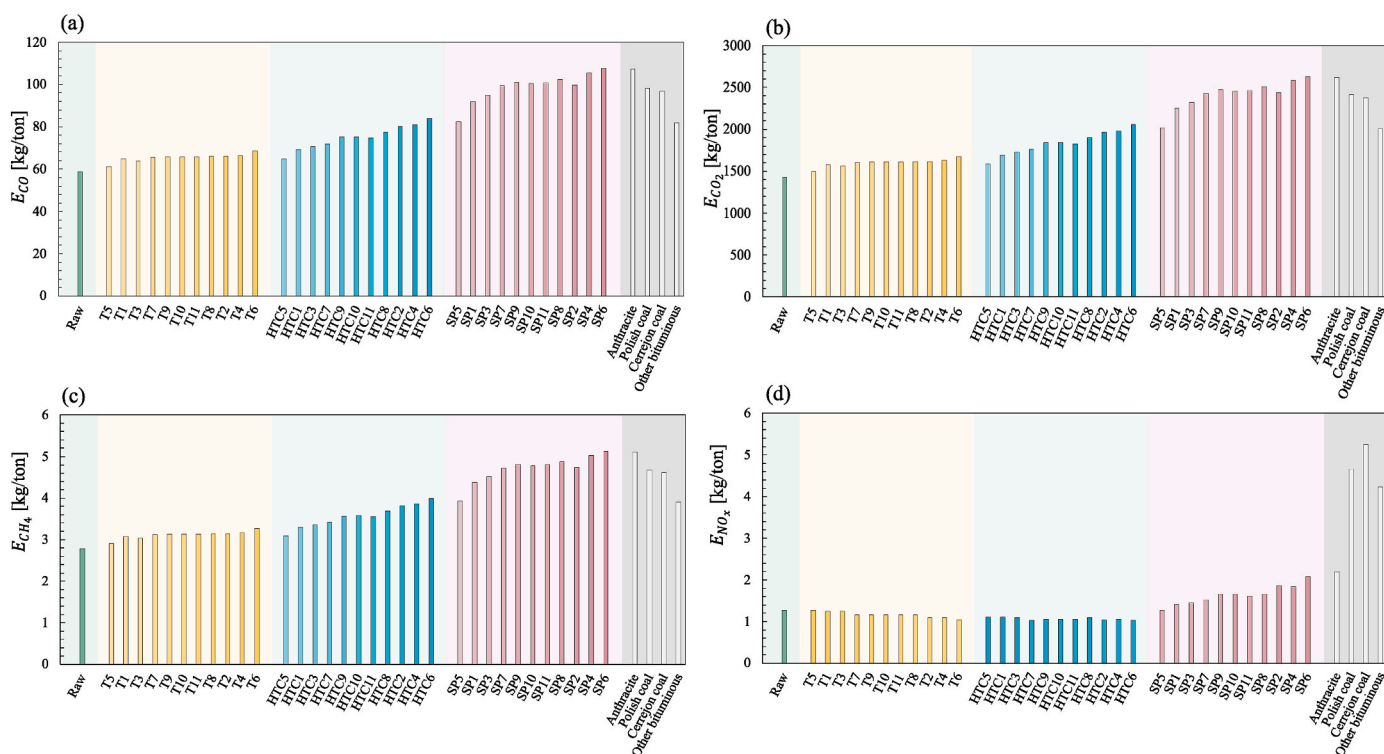


Fig. 8. Emission factors estimation for torrefaction, hydrothermal, and slow pyrolysis treatments. (a) carbon monoxide (E_{CO}), (b) carbon dioxide (E_{CO_2}), (c) methane (E_{CH_4}), and nitrogen oxides (E_{NO_x}) in kg per ton. Data for comparison: anthracite [36], Polish and Cerrejón coal [37], and other bituminous [38].

purposes. Since products distribution depends strongly on process conditions, these can be tailored to prioritize solids yield over liquid or gaseous fractions. Biochar generated from these methods can be used for industrial power generation or in residential combustion units. In particular, torrefied EWR co-firing has demonstrated the ability to maintain boiler efficiencies comparable to coal combustion [40], while reducing emissions associated with global warming, human toxicity and other environmental impacts [41]. Similar, hydrothermally treated biomass has shown greenhouse gas mitigation potential when co-fired with coal [42].

Among the evaluated routes, pyrolytic biochar, characterized by higher FC content and enhanced thermal stability, appears especially suitable for metallurgical applications. Its coal-like properties make it a suitable renewable substitute for fossil-based carbon sources in blast furnace injection and iron ore sintering. Its use can contribute to reducing CO₂ emissions and promoting sustainable practices in the metallurgical sector. As discussed in previous sections, high FC biochars may enable significant coke substitution while maintaining acceptable sintering performance. In mini blast furnaces operation, near complete replacement has been technically proposed for sufficiently carbonized biochars.

The CO₂ emitted by the combustion of eucalyptus-derived biochar could be reabsorbed by eucalyptus forests through photosynthesis, provided sustainable management and continuous replanting are maintained. In Brazil, eucalyptus plantations for energy are periodically renewed, contributing to a potentially negative carbon balance. Additionally, a significant fraction of the trees' biomass, including roots, leaves and bark, remain in the soil after harvest, helping with long-term carbon sequestration.

While the present study provides systematic comparison of TOR, HTC, and SP applied to *Eucalyptus* residues, it primarily focus of solid fuel characteristics. Future research should include detailed evaluation of liquid and gaseous fraction to obtain a more comprehensive understanding of mass and energy distributions. Techniques such as Fourier-transform infrared spectroscopy (FTIR), X-ray diffraction (XRD), and scanning electron microscopy (SEM) would allow for the identification of changes in functional and structural groups, as well as a detailed morphological examination. Furthermore, more detailed inorganic analysis, such as inductively coupled plasma (ICP), would allow the determination of the inorganic elements content present in the samples, making it possible to assess potential slagging and fouling tendencies of the biochar and hydrochar when used in combustion or gasification systems. These analyses would provide a more comprehensive understanding of the material and further enhance its potential applications.

Although all three thermochemical routes improve biomass properties to varying degrees, the selection of a specific technology ultimately depends on intended application, fuel quality targets, costs considerations, and sustainability criteria. This flexibility enables businesses and consumers to tailor biomass conversion pathways to produce customized product aligned with their energy, environmental, and economic goals. Future research should therefore focus on integrating these technologies and the optimized parameters identified in this study into sustainable energy systems for the effective utilization of *Eucalyptus* wood residues. In parallel, life cycle assessment (LCA) would be essential to evaluate the environmental impacts of each conversion route and to ensure their feasibility and sustainability under real-world conditions.

5. Conclusions

The comparative evaluation of torrefaction, slow pyrolysis, and hydrothermal carbonization demonstrated distinct impacts on the physicochemical properties of biochars derived from *Eucalyptus* wood residues. Among the investigated routes, slow pyrolysis produced the most energy-dense and carbon-rich biochar, reaching fixed carbon contents of up to 85% at operational conditions of 662 °C and 4.5 h. These characteristics closely resemble those of high-rank coals,

highlighting the suitability of pyrolytic biochar for industrial applications such as metallurgical fuel substitution. However, this came at the cost of lower solid mass yields, which decreased to approximately 30 %.

Hydrothermal carbonization provided a different balance between fuel upgrading and material retention. Under the most severe conditions evaluated (252 °C and 4.5 h), hydrochar retained around 50 % of the initial mass while achieving moderate improvements in energy properties, such as an increase of about 31 % in higher heating value. This combination makes HTC-derived solids particularly attractive for applications where high yields are prioritized, such as soil amendment or co-firing. Torrefaction showed strong dependence on residence time, with relatively short treatment durations already producing biochar with enhanced energy qualities, demonstrating its potential as an efficient pretreatment option.

Thermogravimetric analysis confirmed the high thermal stability of pyrolytic biochar, with its main degradation stage taking place at temperatures above 600 °C. Additionally, theoretical emission factor estimations indicate combustion-related CO₂ emissions of approximately 2600 kg_{CO2} per ton of biochar, comparable to some coal emissions but of biogenic origin in this case. The distribution of fuel ratio and bioenergy content further highlighted the enhanced combustion quality and energy potential of pyrolytic biochar relative to torrefied and hydrothermally carbonized materials. Importantly, the optimization via response surface methodology revealed that meaningful energy enhancements could be achieved at relatively mild operative conditions, around 180 °C and 20 min for torrefaction and 191 °C and 2.3 h for hydrothermal carbonization, supporting improved process efficiency.

Altogether, the findings demonstrate that *Eucalyptus* wood residues constitute a viable feedstock for producing solid biofuels with properties tailored to specific end-uses. Selecting an appropriate conversion technology should consider trade-offs between mass yield, energy content, and application requirements. Thus, the insights from this study support the strategic valorization of eucalyptus residues through thermochemical processing, contributing to sustainable energy systems and reducing dependence on fossil-derived carbon.

CRedit authorship contribution statement

Orlando Salcedo-Puerto: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Clara Mendoza-Martinez:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Thiago de Paula Protásio:** Writing – review & editing, Supervision, Data curation, Conceptualization. **Esa Vakkilainen:** Writing – review & editing, Supervision, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Thiago de Paula Protasio reports financial support was provided by Minas Gerais State Foundation of Support to the Research (APQ-05431-24, grant ID FAPEMIG/DPE n°. 103658065/2024). Thiago de Paula Protasio reports financial support was provided by National Council for Scientific and Technological Development (grant ID 304165/2022-0). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.renene.2026.125545>.

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